Midterm Project

Background:

In modern software development, **database management** and **web server deployment** are fundamental skills. Professional programmers frequently work with SQL databases such as **Azure SQL or AWS SQL** and configure software on various server environments.

This project provides hands-on experience in **data acquisition, storage, and retrieval**, reflecting real-world development challenges. Through this project, you will demonstrate proficiency in:

- **Data Acquisition & Storage**: Extract and store data from a web-based input form.
- **Database Management**: Set up and manage an **SQLite** database.
- **Web Server Implementation**: Use a **Python server** to process and store data.
- **Data Retrieval & Export**: Retrieve stored data and export it as a **CSV file**.

Objective:

This project is designed to **simulate real-world problem-solving** and help you build **industry-relevant** skills. Instead of focusing solely on technical implementation, emphasize **understanding the overall workflow**—from data entry to retrieval and export.

Your task is to **develop a web-based system** that connects a **website to a database**, storing user input, retrieving it upon request, and exporting it as a **CSV file**.

Project Requirements

1. Database Setup

- Install **SQLite3.exe** from the provided folder on your **Windows/macOS** system.
- Create a **database table** to store user-submitted data.

2. Web Interface

- Set up a **local website** using the **provided HTML template**.
- The form should contain **two input fields** and a **submit button**.
- You are welcome to use css to make it pretty

3. Server-Side Processing

- Implement a **Python-based server** to handle form submissions.
- Store the **submitted data** in the SQLite database.
- Enable **data retrieval** and **export** to a **CSV file**.

4. Provided Resources

- **HTML template** with comments for guidance.
- **SQLite installation instructions** for Windows and macOS.
- **Python server template** for handling database interactions.
- **Sample user input data** (see below).

Sample Input from the Website

Name: Alice Smith, 4-digit-Phone: 1234
Name: Bob Johnson, 4-digit-Phone: 1230
Name: Charlie Williams, 4-digit-Phone: 5678
Name: David Brown, 4-digit-Phone: 0912
Name: Eve Davis, 4-digit-Phone: 3456
Name: Frank Miller, 4-digit-Phone: 7890
Name: Grace Wilson, 4-digit-Phone: 2345
Name: Henry Moore, 4-digit-Phone: 6789
Name: Ivy Taylor, 4-digit-Phone: 0003
Name: Jack Anderson, 4-digit-Phone: 8765

Expected Learning Outcomes

By completing this project, you will:
- Understand how **web applications interact with databases**.
- Learn how to **design and implement a database** for structured data storage.
- Gain experience in **data retrieval and CSV file exports**.
- Apply **Python and SQL** together in a **real-world** development scenario.

Definition of Done

- The deliverable for the project is a ZIP file containing all of the files in your working folder. Submit the ZIP file in the midterm project folder in Blackboard.
- Your instructor should be able to start your web server and use the project web page to add data to your table. He or she should also be able to click the CSV link on your web page and view all of the data in your table in CSV format.